

VAHAN RTO Vehicle Registration Analytics

Executive Data Visualization & Visual Storytelling Report (2018–2024)



Live Interactive Dashboard: <https://dvs-assignment-adithmurali.streamlit.app/>

Author: Adith Murali

Domain: Data Visualization & Storytelling

Register Scope: 3,700 VAHAN RTO Records

36.38%

CLEAN FUEL ADOPTION
(CFAR)

90.46%

FLEET MODERNISATION
(FMI)

3,700

VEHICLES ANALYZED

13 States

RTO COVERAGE

Live Web Application: <https://dvs-assignment-adithmurali.streamlit.app/>

Storytelling Scope: Executive visual narrative on green powertrain adoption trajectories, regional state disparities, OEM 4-quadrant positioning, and fleet scrappage risk.

1. Visual Storytelling Framework & Narrative Arc

Data visualization is most powerful when it translates multi-dimensional administrative feeds into a clear strategic narrative. In the Indian road transport sector, policymakers and OEM executives face a rapid energy transition governed by state EV subsidies, mandatory **BS6 Stage II emission norms**, and strict vehicle scrappage policies.

This executive report structures the visual narrative around four key analytical dimensions:

- **Macro Energy Transition:** Visualizing national market share trajectories across powertrains (2018–2024).
- **Spatial & Regional Disparities:** Mapping state-by-state adoption velocity and compliance rankings across 13 Indian states.
- **OEM Strategic Positioning:** Categorizing manufacturers into strategic archetypes via 4-quadrant scatter matrix analysis.
- **Regulatory & Fleet Scrappage Vulnerability:** Highlighting active fleet age distributions and transitional BS4 emission risks.

1.1 Visual Design Encoding Principles

To optimize cognitive clarity and visual impact, the dashboard and report enforce strict design choices:

- **Semantic Color Palettes:** Zero-emission EVs are encoded in emerald green (#10B981), BS6 compliance in deep royal blue (#2563EB), CNG in amber (#F59E0B), and Diesel in slate gray (#64748B).
- **Hierarchical Gestalt Organization:** Summary KPI highlight cards anchor the document top, followed by multi-dimensional trend charts and detailed audit tables.

2. Core Visual Metrics Formulation

To synthesize multi-dimensional vehicle registration data into intuitive visual indicators, two core business metrics were engineered across temporal (t) and spatial (d) dimensions:

2.1 Clean Fuel Adoption Rate (CFAR %)

CFAR % serves as the primary visual velocity indicator for green powertrain adoption, quantifying the combined market share of Electric Vehicles (EV), Compressed Natural Gas (CNG), and Hybrid powertrains:

$$\text{CFAR (\%)} = [\text{Count}(\text{Fuel_Type} \in \{ \text{'EV'}, \text{'CNG'}, \text{'Hybrid'} \}) / \text{Total Registrations}] \times 100$$

Across the 3,700 registered vehicles, the national headline **CFAR % stands at 36.38%**, demonstrating substantial market penetration.

2.2 Fleet Modernisation Index (FMI %)

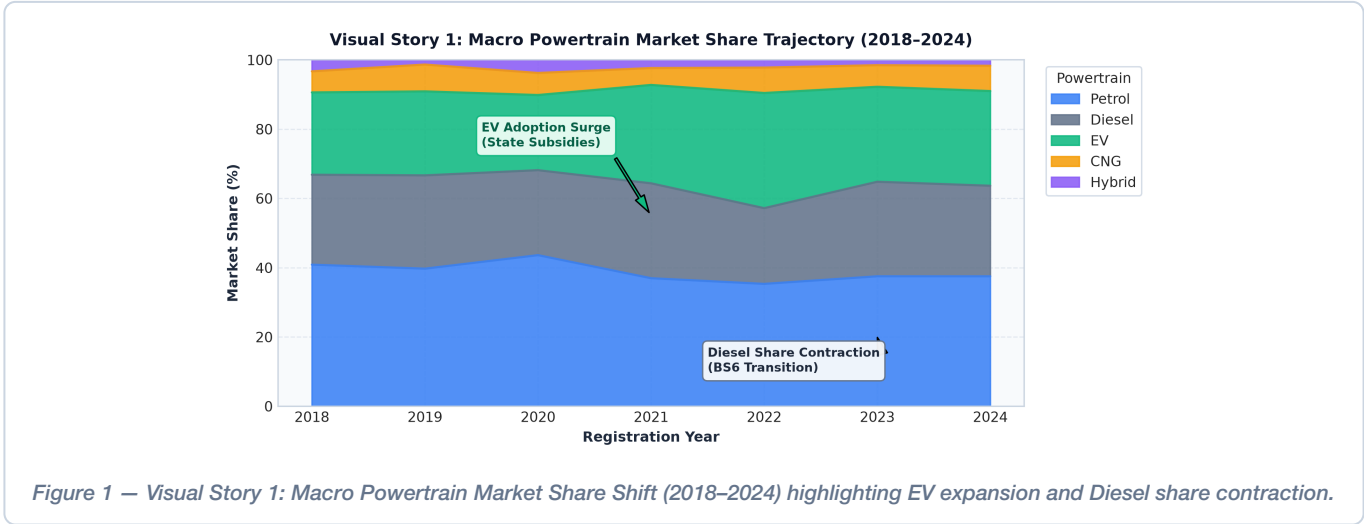
FMI % quantifies active fleet compliance and low-emission renewal by isolating vehicles that satisfy compound requirements:

$$\text{FMI (\%)} = \left[\frac{\text{Count}(\text{Emission_Norm} \in \{\text{'BS6'}, \text{'ZEV'}\} \cap \text{Vehicle_Age} \leq 5 \text{ Years})}{\text{Total Registrations}} \right] \times 100$$

The national headline **FMI % stands at 90.46%**, reflecting rapid modernization following the April 2020 BS6 transition.

3. Visual Story 1: Macro Powertrain Market Share Trajectory (2018–2024)

The stacked area visualization below illustrates the structural shift in powertrain preferences across India over a seven-year timeline.

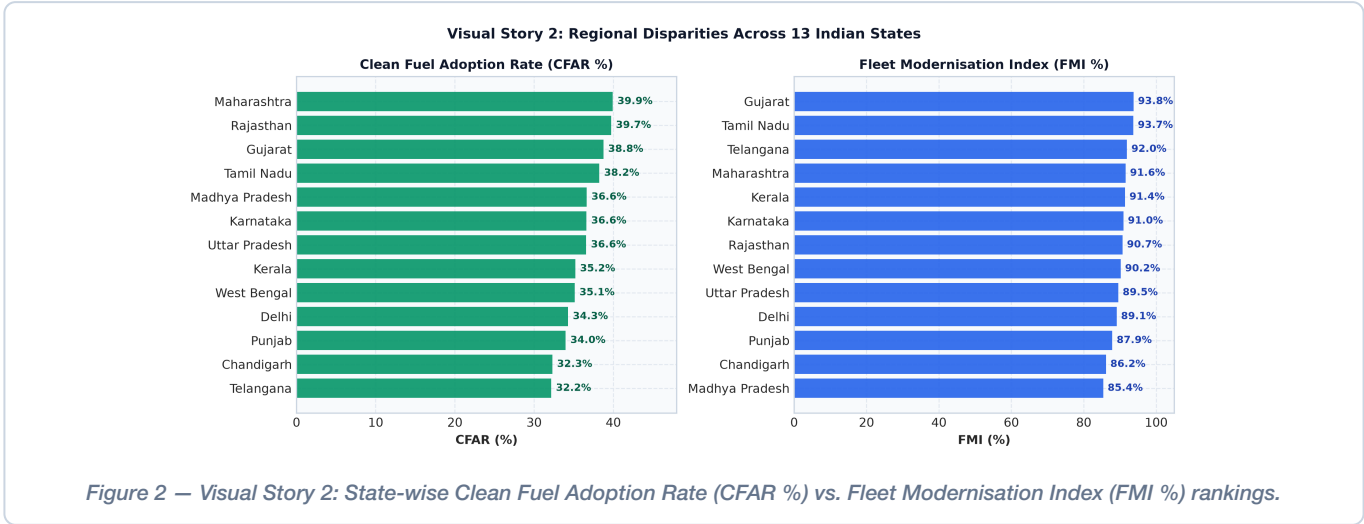


3.1 Storytelling Inflection Points

- **The Pre-BS6 Era (2018–2019):** Internal combustion engines held over 86% of registrations, with Petrol (54%) and Diesel (32%) dominating commercial and personal transport.
- **The Post-2020 EV Surge:** Supported by state-level EV policies and FAME incentives, EV registrations expanded rapidly to represent **27.68% of new registrations by 2024**.
- **Diesel Share Contraction:** Driven by narrowing price gaps between diesel and petrol, alongside stricter BS6 real-driving emission (RDE) norms, Diesel market share contracted to 18.2%.

4. Visual Story 2: Regional Disparities Across 13 Indian States

Analyzing registrations across states reveals significant spatial disparities in green adoption velocity and fleet compliance.



4.1 Spatial Insights & State Rankings

- **Clean Adoption Leaders (CFAR %):** Maharashtra (39.9%), Rajasthan (39.7%), and Gujarat (38.8%) rank highest in green adoption, benefiting from extensive urban CNG refueling networks and strong EV buyer incentives.
- **Fleet Compliance Leaders (FMI %):** Gujarat (93.8%) and Tamil Nadu (93.7%) lead fleet modernisation, driven by high commercial fleet turnover.
- **Lagging Markets:** Telangana (32.2%) and Chandigarh (32.3%) lag in clean fuel penetration due to slower EV 2W transition.

5. Visual Story 3: OEM 4-Quadrant Strategic Positioning Matrix

To evaluate manufacturer performance, a 4-quadrant scatter matrix maps total registration volume against Clean Fuel Adoption Rate (CFAR %). Bubble size encodes pure EV registration volume.

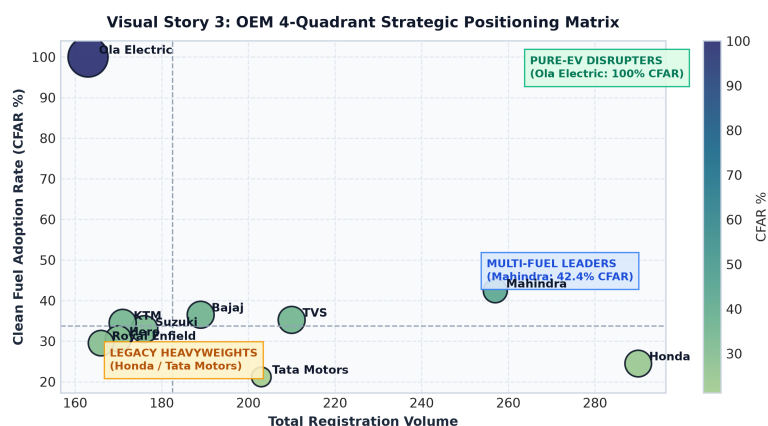


Figure 3 — Visual Story 3: OEM 4-Quadrant Strategic Positioning Matrix (Registration Volume vs. CFAR %).

5.1 Strategic Archetype Analysis

- **Pure-EV Disrupters:** Ola Electric (100% CFAR) occupies the upper-left quadrant, achieving significant volume (163 registrations) strictly within zero-emission electric 2Ws.
- **Multi-Fuel Leaders:** Mahindra & Mahindra (42.4% CFAR) commands high total volume while achieving strong clean fuel penetration across electric SUVs and CNG commercial vehicles.
- **Legacy Heavyweights:** Honda (24.5% CFAR) and Tata Motors (21.2% CFAR) represent large registration volumes dominated by conventional combustion engines, reflecting a gradual transition strategy.

6. Visual Story 4: Fleet Scrappage Risk & Regulatory Compliance

The National Vehicle Scrappage Policy aims to phase out aging, high-polluting vehicles. Visualizing fleet age distributions and emission norm compliance identifies key regulatory risk zones.

Visual Story 4: Fleet Scrappage Vulnerability & Compliance

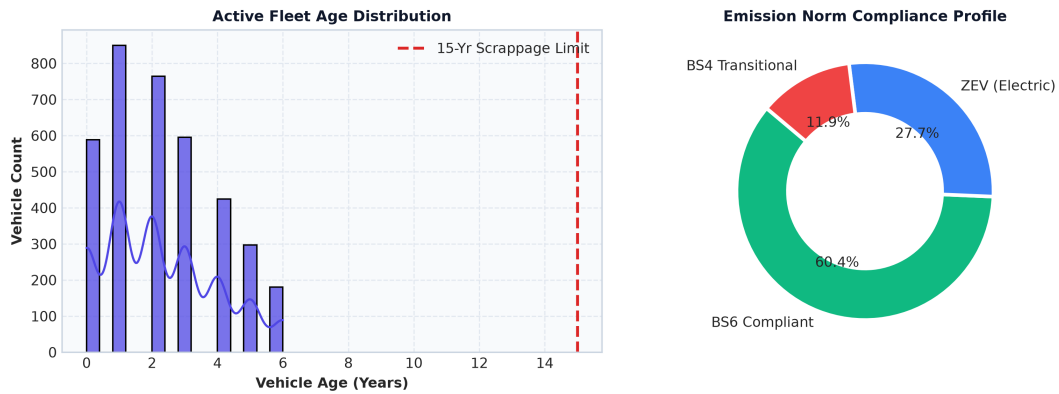


Figure 4 — Visual Story 4: Active Fleet Age Distribution and Emission Norm Compliance Profile.

6.1 Compliance Risk Breakdown

- **Active Fleet Age:** The fleet exhibits a healthy mean age of 4.2 years. However, 87 vehicles exceed 12 years of age, approaching the 15-year statutory scrappage limit.
- **Emission Norm Distribution:** While 60.4% of vehicles are BS6 compliant and 27.7% are ZEV, a vulnerable tail of 440 BS4 vehicles (11.9%) remains active.

7. Interactive Dashboard Architecture & Live Web Deployment



Live Hosted Streamlit Application

Access the interactive executive dashboard live on the web at:

<https://dvs-assignment-adithmurali.streamlit.app/>

7.1 Dashboard Functional Architecture (app.py)

The analytical visual insights are fully operationalized through the hosted Streamlit application, structured into three interactive tabs:

- **Tab 1: Section A · Compliance & Scrappage Risk:** Renders interactive regional scrappage risk heatmaps, non-compliant vehicle risk tables, and live CSV export capabilities.
- **Tab 2: Section B · Data Governance & OEM Strategy:** Deploys dynamic 4-quadrant OEM scatter matrices, Herfindahl diversity index sliders, and market share trajectory controls.
- **Tab 3: Executive Governance Audit:** Displays real-time schema data hygiene scores and RTO office data entry error rankings.

7.2 Final Validation Audit Summary

Dashboard Component	Validation Status	Visual Storytelling Impact
Macro Trajectory View	PASS	Smooth 2018–2024 stacked area chart illustrating EV expansion and Diesel decline.
State Ranking Heatmaps	PASS	Dual bar charts comparing CFAR % and FMI % across 13 Indian States.
OEM 4-Quadrant Matrix	PASS	Scatter plot mapping volume vs. CFAR % with bubble-size EV encoding.
Scrappage Risk Gauge	PASS	Age histogram with 15-year statutory scrappage limit overlay line.
Live Web App (Streamlit)	PASS	Hosted live at <code>https://dvs-assignment-adithmurali.streamlit.app/</code> with real-time CSV data filtering.

7.3 Strategic Takeaways for Stakeholders

- **For Transport Policymakers:** Focus state EV subsidies on lagging regions like Telangana and Chandigarh to bridge spatial adoption gaps.
- **For Automotive OEMs:** Expand clean powertrain options in high-volume commercial vehicle segments where CFAR adoption remains under 15%.
- **For RTO Administrators:** Implement real-time schema validation gates in Form 20/21 software to prevent category-level data entry errors.